

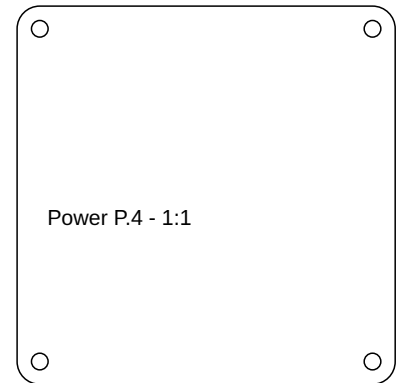
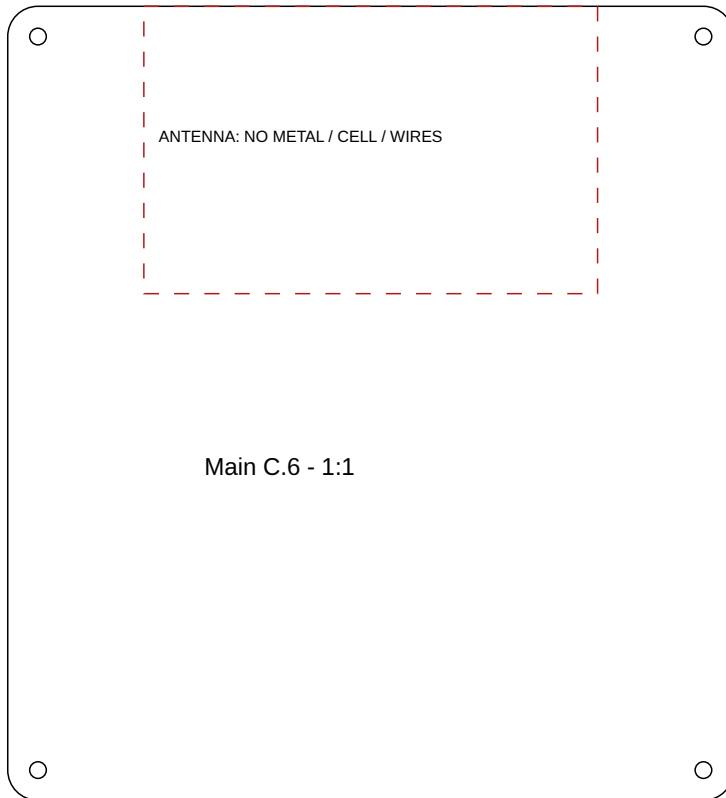
Keychain Kreatures - separate mounting templates

C.6 / P.4 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

Main C.6: 96 x 105 mm, R4. Power P.4: 50 x 50 mm, R3. Both 1.6 mm thick.

Each board has four 2.2 mm M2 holes. The mounting patterns DO NOT MATCH.

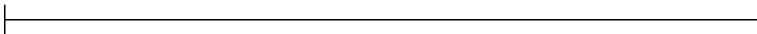
Coordinates: each board has its own top-left origin, X right, Y down.



Main holes: (4,4), (92,4),
(4,101), (92,101) mm.

Power holes: (3,3), (47,3),
(3,47), (47,47) mm.

100 mm calibration line:



PRINT AT 100% / ACTUAL SIZE. Disable fit-to-page. Measure the line first.

Main display/buttons face the user. Smaller power PCB fits behind on its own supports.

These templates show individual board geometry, NOT an approved stack placement.

Keep the entire power PCB, cell, harness and supports clear of the main antenna zone.

There is no shared four-standoff mounting pattern or approved board spacing.

Check actual components, plugged JST bodies, wire bends and solder tails on both sides.

Clip solder tails to ≤ 2.5 mm, inspect every joint, and keep loose wires restrained.

Unscrew and unplug the rear power board to expose the student main-board parts.

Do not trap a battery between solder tails and components. Cell position is not frozen.

Keep a protective insulating barrier; never use battery foil as an electrical insulator.

Flat parts and assembly limits

C.6 / P.4 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

R1-R43

Existing Yageo MFR-25 values

All horizontal, 10.16 mm lead pitch

C1,C3,C5,C13,C17,C25

Panasonic ECEA1CKA101

100 uF / 16 V; max flat height 7.3 mm*

C9,C11,C15,C21

Panasonic ECEA1CKA100

10 uF / 16 V; max flat height 5.0 mm*

Q1-Q6

Existing exact transistor MPNs

Flat marked face up; preserve pin order

* Manufacturer maximum can diameter plus 0.5 mm insulating support; not scaled artwork.

Both capacitor types use the full 8 mm maximum body-length courtyard.

Capacitor square pad 1 is POSITIVE. Sleeve stripe marks NEGATIVE pad 2.

10 uF native lead pitch is 1.5 mm: form to the 2.54 mm PCB pitch without crossing.

Support leads at the seal while bending; do not pull, twist, or stress the rubber seal.

Use nonconductive support under horizontal cans. Do not obstruct the pressure vent.

TO-92 bodies extend toward the footprint body rectangle; marked flat face stays up.

Q6 is LP0701N3-G, not BC327. Verify actual pinout before forming or soldering.

IR lenses must still face the case opening; optical parts are not flattened blindly.

Sockets, screen, buttons and the front RGB retain their actual functional height.

The socketed ESP32 rear envelope is about 15.3 mm; it, not the caps, limits spacing.

The socketed RGB front envelope remains about 22.1 mm. Case depth is NOT final.

Power board remains a factory-assembled SMT subassembly, not a student SMT kit.

Prototype checks still required

Measure delivered screen, ESP32, SD module, sockets, plugs and standoffs.

Check horizontal capacitor ripple/rail transients during Wi-Fi, audio and motor bursts.

Check component temperatures, charging, rail sequencing and battery protection.

Confirm antenna performance with the actual rear board, battery, harness and case.

Prototype files issued separately; approve physical fit and factory DFM before ordering.

Main-board rear - populated mechanical preview

C.6 / P.4 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

Preview is not a powered test. Seller-module envelopes and formed leads need sample checks.

Current projects: KK_main_module (C.6) and KK_power_module (P.4).

Compact power board - 50 x 50 mm, separate supports

C.6 / P.4 PROTOTYPE REVIEW - PHYSICAL / BENCH QUALIFICATION PENDING

Routed power core preserved. USB at top and switch at right in this native front view.

Power J3 -> main J1: 1=5V, 2=GND, 3=3.3V, 4=3.2V. Keyed pin-to-pin harness.

Case placement and board spacing are NOT frozen. Never overlap the main antenna zone.